MLP

Linear (784, 256)

input and output sizes:

$N_{in} = 784$ bias: one bias per output

$N_{out} = 256$

$N\ params = N_{in} \cdot N_{out} + N_{out}$

$N\ params = 784 \cdot 256 + 256 = 200,704 + 256 = 200,960$

$Macs = N_{in} \cdot N_{out}$

$Macs = 784 \cdot 256 = 200704$

$Flops = 2 \cdot Macs$

$Flops = 2 \cdot 200704 = 401408$

Results of Linear (784 256)

Macs	200 704
Flops	401 408

CNN

Conv2d (16, 32, kernel size = 3, padding = 1)

Input size

$H_{in} = 14$, $W_{in} = 14$, $C_{in} = 16$

Kernel size $K = 3 \times 3$ Output channels $C_{out} = 32$

Padding = 1, Stride = 1 Output size: $H_{out} = 14 = W_{out} = 14$

$N_{params} = C_{out} \cdot (C_{in} \cdot K \cdot K + 1)$

$N_{params} = 32 \cdot (16 \cdot 3 \cdot 3 + 1) = 32 \cdot (144 + 1) = 32 \cdot 145$

$= 4640$

$Macs = H_{out} \cdot W_{out} \cdot C_{out} \cdot C_{in} \cdot K \cdot K$

$Macs = 14 \cdot 14 \cdot 32 \cdot 16 \cdot 3 \cdot 3 = 196 \cdot 32 \cdot 16 \cdot 9 = 903\,168$

$Flops = 2 \cdot Macs$

$Flops = 2 \cdot 903\,168 = 1\,806\,336$

~~Model~~ Multi Head Attention

Model parameters

number of pathes: $S = 16$ Embedding dimension: $d = 64$

number of heads: $h = 4$ Ffn hidden dimension: $dff = 128$

Each $S \cdot d \cdot d \rightarrow 3 \cdot S \cdot d^2$ in total

$Macs\ QKV = 3 \cdot d^2 = 3 \cdot 16 \cdot 64^2 = 196\,608$

$Flops\ QKV = 2 \cdot 196\,608 = 393\,216$

$Macs\ QKT = S^2 \cdot d = 16^2 \cdot 64 = 16\,384$

$Flops\ AV = 2 \cdot 16\,384 = 32\,768$

$Macs\ O = S \cdot d^2 = 16 \cdot 64^2 = 65,536$

$Flops\ O = 2 \cdot 65,536 = 131\,072$

$MACs\ FFN = 2 \cdot S \cdot d \cdot dff = 2 \cdot 16 \cdot 64 \cdot 128 = 262,144$

$FLOPs\ FFN = 2 \cdot 262\,144 = 524\,288$

Results

$MACs = 294\,912 + 262\,144 = 557\,056$

$Flops = 1\,114\,112$